Arista Access Node MG-ACC-38 SOP for Hardware/Environmental Fault (ASIC memory parity errors, line card crashes, power supply redundancy loss)

Node: MG-ACC-38 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_9755

Description

This Standard Operating Procedure (SOP) outlines the steps to troubleshoot and resolve hardware/environmental faults (ASIC memory parity errors, line card crashes, power supply redundancy loss) on Arista Access Node MG-ACC-38 at MG Road Core. The Access Node serves as the entry point into the network, providing Layer 2 connectivity to customer premises equipment (CPE). The Arista EOS (Extensible Operating System) running on the node enables features like Spanning Tree Protocol, VLANs, and trunking. However, in the event of hardware/environmental faults, the node's ability to operate and forward traffic is compromised, resulting in critical network downtime. ASIC memory parity errors and line card crashes are hardware-related issues that can occur due to manufacturing defects, environmental factors like temperature and humidity fluctuations, or power supply issues. The loss of power supply redundancy is a reliability concern, as it can cause the node to fail, leading to network unavailability. The root causes of these issues are often interrelated, making comprehensive troubleshooting and diagnostic steps essential. This SOP provides a detailed procedure to isolate and resolve the root cause of the fault, ensuring minimum network downtime and optimal network performance.

Pre-requisite Checks

- check: Verify power supply redundancy status using the command: `show system redundancy`, description: This command checks the power supply redundancy status, ensuring that there are no issues with the power supply units (PSUs).

- check: Confirm redundant link status using the command: `show interfaces status`, description: This command verifies the status of redundant links, ensuring that there are no issues with the link interfaces.

- check: Dispatch remote hands to MG Road Core for any necessary physical checks, description: Remote hands may be required to perform physical checks, such as inspecting the node's hardware, checking for signs of environmental damage, or verifying power supply connection.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system redundancy, Expected Output: Redundancy status: Active/Standby, Description: This command displays the system redundancy status, showing whether the node is operating in active or standby mode. A standby node is not currently operating but is ready to take over in case the active node fails. The command helps identify potential issues with the power supply redundancy.

Command: show interfaces status, Expected Output: Interface Status: up/down, Description: This command displays the status of each interface, including the redundant links. A down interface indicates a potential issue with the link or the node.

Command: show system resources memory, Expected Output: Memory usage: 90%, Description: This command displays the system memory usage, helping identify potential issues with memory allocation or usage.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC, Command: replace system asic, Description: If the diagnosis indicates a faulty ASIC, replace it with a new one. The command `replace system asic` initiates the process of replacing the ASIC.

Step 2: Step: Reboot the node, Command: reload, Description: After replacing the ASIC, reboot the node to ensure that the new component is recognized and functioning correctly. The `reload` command initiates the reboot process.

Step 3: Step: Verify power supply redundancy status, Command: show system redundancy, Description: After resolving the issue, verify that the power supply redundancy status is healthy. The command `show system redundancy` confirms that the node is operating with redundant power supplies.

Post-Check Verification

```
Command: show system resources memory, Expected Output: Memory usage: 90%, Description: This command verifies that the memory usage has returned to normal levels, indicating that the issue has been resolved.

Command: show interfaces status, Expected Output: Interface Status: up/up, Description: This command verifies that all interfaces, including the redundant links, are operating correctly.
```

Roll-back Procedures

- Step: Return the node to its original configuration, Command: copy running-config startup-config, Description: If the fix fails, return the node to its original configuration using the `copy running-config startup-config` command.

- Step: Revert the changes made during the resolution process, Command: rollback asic upgrade, Description: Revert the changes made during the resolution process, including any ASIC upgrades or replacements, using the `rollback asic upgrade` command.